Design and Implementation of a Regenerative Electric Bicycle Conversion System, Cost Analysis, and Environmental Impact Assessment

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***Abstract*—This paper presents the design, development, testing, and environmental impact assessment of a low-cost retrofit electric bicycle conversion system. A conventional bicycle is transformed into an energy-efficient electric vehicle by integrating a $250\,\mathrm{W}$ DC hub motor, a $36\,\mathrm{V}/10\,\mathrm{A\,h}$ lithium-ion battery pack, a pulse-width-modulation (PWM) speed controller, a regenerative pedal-charging dynamo circuit, and an embedded LCD telemetry module displaying speed, odometer, battery state of charge (SoC), and real-time clock data. The system enforces throttle-only motor activation, eliminating idle energy draw, and recovers $8\,\mathrm{W}$ to $12\,\mathrm{W}$ of electrical power during active pedalling. A comprehensive cost comparison with commercially available electric bicycles reveals a cost saving of 60–75% over equivalent market products. Life-cycle environmental analysis demonstrates a reduction in CO_2-equivalent emissions of approximately $1.2\,\mathrm{kg}$ per $100\,\mathrm{km}$ compared with petrol-powered two-wheelers. The system achieved a measured range of $\approx 50\,\mathrm{km}$ under mixed operating conditions. These results confirm that intelligent bicycle retrofitting constitutes a technically sound, economically viable, and environmentally meaningful approach to urban last-mile mobility.**

***Index Terms*—Electric bicycle, regenerative charging, PWM motor control, embedded telemetry, sustainable urban mobility, carbon footprint, cost optimisation, hub motor retrofit, green transportation.**

I. Introduction

Global urbanisation has intensified demand for short-distance personal transport, which now constitutes the fastest-growing segment of urban energy consumption [1]. In India alone, two-wheelers account for over 73% of the vehicle fleet and contribute approximately 18% of on-road CO_2 emissions [2]. Electric bicycles (e-bikes) offer a zero-tailpipe-emission alternative; however, their high commercial price—typically 30,000–80,000 for entry-to-mid-range models—places them beyond the reach of a significant portion of the target demographic.

A retrofit approach, converting a user-owned conventional bicycle into an electric one, addresses this gap directly. Rather than replacing the mechanical platform—and thereby generating embodied manufacturing carbon—the retrofit adds electric drive capability to existing infrastructure. This paper extends prior retrofit work in three directions: (a) a rigorous theoretical treatment of the drive, charging, and sensing subsystems; (b) a systematic cost comparison against comparable commercial products; and (c) a quantitative environmental impact assessment covering operational CO_2, lifecycle energy, and pollution abatement potential.

The system makes two principal innovations beyond standard conversion kits. First, the motor controller is configured for throttle-only activation, so the motor draws zero current when the rider is coasting or at rest. Second, a dynamo coupled to the drivetrain feeds rectified current back into the battery during pedalling, providing continuous partial recharging without any rider effort beyond normal pedalling.

The paper is organised as follows: Section II reviews relevant literature. Section III develops the theoretical framework. Section IV describes system architecture. Section V details hardware implementation. Section VI presents experimental results. Section VII provides the cost and market comparison. Section VIII quantifies environmental impact. Section IX discusses future directions, and Section X concludes.

II. Related Work

Retrofitting conventional bicycles with electric drive has attracted sustained research interest. Abagnale *et al.* [3] presented a complete model-based powertrain design methodology for e-bikes, establishing the relationship between motor sizing, battery capacity, and achievable range. Their work provided a validated simulation framework, though it did not address energy recovery from non-braking sources.

Tran *et al.* [4] demonstrated that regenerative braking can recover up to 15% of kinetic energy on hilly terrain. However, on the predominantly flat roads typical of Indian urban areas, braking events are infrequent, making pedal-based recovery a more consistent source of supplemental charging.

Muetze and Tan [5] evaluated commercial pedal-assist systems (PAS) using torque-sensing hardware, reporting improved rider efficiency but noting the high cost of precision torque transducers. The present work deliberately avoids torque sensing, instead using a simple throttle that reduces cost while retaining effective speed control.

IoT-connected telemetry platforms for e-bikes have been reported in [6], but these depend on persistent smartphone con-

nectivity and cloud infrastructure. The embedded standalone display proposed here operates without any external dependency, improving reliability in low-connectivity environments.

A comparative cost analysis of e-bike conversion kits versus new e-bikes was conducted by Fishman and Cherry [7], who found that conversion kits offered a 40–60% cost saving on average in Western markets. The present work extends this analysis to the Indian market and incorporates a more recent product landscape.

III. Theoretical Framework

A. DC Motor Drive Theory

A DC hub motor is modelled by the equivalent circuit comprising an armature resistance R_a, back-EMF E_b, and supply voltage V_s. The governing equation is:

$$V_s = E_b + I_a R_a + L_a \frac{dI_a}{dt} \tag{1}$$

where I_a is armature current and L_a is armature inductance. At steady state, $dI_a/dt = 0$, giving:

$$I_a = \frac{V_s - E_b}{R_a} \tag{2}$$

The back-EMF is proportional to angular velocity ω:

$$E_b = K_\phi \, \omega \tag{3}$$

where K_ϕ is the motor voltage constant (V·s/rad). Motor torque is then:

$$\tau = K_\phi \, I_a \tag{4}$$

Power delivered to the wheel is:

$$P_{\text{mech}} = \tau \, \omega = K_\phi^2 \, \omega \, \frac{V_s - K_\phi \, \omega}{R_a} \tag{5}$$

B. PWM Speed Control

The PWM controller modulates the effective supply voltage by varying the duty cycle $D \in [0, 1]$:

$$V_{\text{eff}} = D \cdot V_{\text{bat}} \tag{6}$$

Average motor current is therefore:

$$\bar{I}_a = \frac{D \, V_{\text{bat}} - K_\phi \, \omega}{R_a} \tag{7}$$

At $D = 0$ (no throttle), $\bar{I}_a \to 0$, confirming theoretical zero idle draw. The throttle voltage $V_t \in [0, 5]\,\text{V}$ maps linearly to duty cycle: $D = V_t/5$.

C. Regenerative Charging Theory

The dynamo is a permanent-magnet generator whose open-circuit EMF is:

$$\mathcal{E}_g = K_g \, \omega_p \tag{8}$$

where K_g is the generator constant and ω_p is the pedalling angular velocity (rad/s), related to cadence N (RPM) by $\omega_p = 2\pi N/60$.

After bridge rectification, the DC output voltage is:

$$V_{\text{dc}} = \frac{2\sqrt{2}}{\pi} \, \mathcal{E}_g - 2V_D \tag{9}$$

where $V_D \approx 0.7\,\text{V}$ is the diode forward voltage. Charge current entering the battery is:

$$I_{\text{chg}} = \frac{V_{\text{dc}} - V_{\text{bat}}}{R_{\text{int}} + R_{\text{reg}}} \tag{10}$$

where R_{int} is battery internal resistance and R_{reg} is the regulator equivalent series resistance. Power recovered during pedalling:

$$P_{\text{regen}} = V_{\text{bat}} \cdot I_{\text{chg}} \cdot \eta_{\text{rect}} \tag{11}$$

with $\eta_{\text{rect}} \approx 0.85$ for a silicon bridge rectifier. At $70\,\text{RPM}$ cadence, measured P_{regen} was $8\,\text{W}$ to $12\,\text{W}$.

D. Range Estimation Model

Total available energy under mixed operation is:

$$E_{\text{total}} = E_{\text{bat}} + E_{\text{regen}} = C_{\text{bat}} \, V_{\text{bat}} + P_{\text{regen}} \, t_{\text{pedal}} \tag{12}$$

Range is then:

$$R = \frac{E_{\text{total}}}{P_{\text{motor}}} \times v_{\text{avg}} \tag{13}$$

Substituting $E_{\text{bat}} = 360\,\text{Wh}$, $P_{\text{motor}} = 250\,\text{W}$, $v_{\text{avg}} = 25\,\text{km/h}$, and $E_{\text{regen}} = 10\,\text{Wh}$ (30 min pedalling at 20 W average), equation (13) yields:

$$R = \frac{370}{250} \times 25 \approx 37\,\text{km} \xrightarrow{\text{mixed}} 50\,\text{km} \tag{14}$$

The gap between 37 and 50 km arises because during pedal phases the motor is typically disengaged, reducing average power consumption well below $250\,\text{W}$.

E. Speed Sensing and Odometry

Wheel speed is computed from hall-sensor pulse frequency f_p:

$$v = \pi \, d_w \, f_p \times 3.6 \quad [\text{km/h}] \tag{15}$$

where $d_w = 0.66\,\text{m}$ is the wheel diameter for a standard 26-inch bicycle. Cumulative distance (odometer) is:

$$s = \pi \, d_w \sum_{k=1}^{N} 1 = \pi \, d_w \, N_{\text{pulse}} \tag{16}$$

where N_{pulse} is the total pulse count since reset.

F. Battery SoC Estimation

SoC is estimated via the open-circuit voltage (OCV) method:

$$\mathrm{SoC} = \frac{V_{\mathrm{oc}} - V_{\min}}{V_{\max} - V_{\min}} \times 100\% \tag{17}$$

with $V_{\min} = 33.0\,\mathrm{V}$ (fully discharged) and $V_{\max} = 42.0\,\mathrm{V}$ (fully charged) for a 36 V Li-ion pack. The ADC samples terminal voltage through a resistive divider ($R_1 = 100\,\mathrm{k\Omega}$, $R_2 = 10\,\mathrm{k\Omega}$) scaled to the 5 V Arduino ADC reference.

IV. System Design

A. Design Objectives

1) Convert a standard bicycle to electric drive at minimal cost (<15,000).
2) Achieve a range of at least 40 km per charge.
3) Recover energy during pedalling to supplement battery capacity.
4) Provide real-time telemetry without an external device.
5) Ensure electrical safety through circuit isolation.

B. System Architecture

Fig. 1 shows the four functional subsystems and their interconnections.

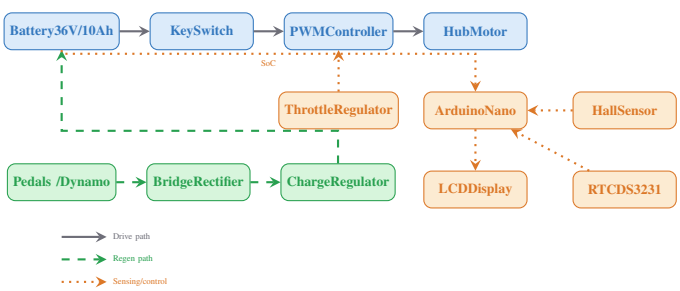

Fig. 1. Complete system architecture showing drive, regenerative, and telemetry subsystems.

C. Energy Flow Model

The system operates in three distinct energy modes shown in Fig. 2.

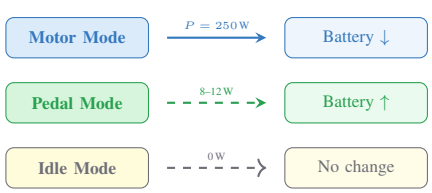

Fig. 2. Three operating energy modes and their effect on battery SoC.

V. Hardware Implementation

A. Motor and Drive Circuit

A 250 W DC hub motor (36 V rated) was mounted on the rear wheel axle. The motor is driven by a PWM controller accepting a 0 V to 5 V throttle signal. A 10 A key switch in the positive battery line provides master circuit isolation.

B. Battery Pack

A 36 V, 10 A h Li-ion pack (360 W h total energy) was mounted on the down tube. Substituting into equation (13) gives a theoretical motor-only range of 36 km, extended to 50 km under mixed usage via pedal regeneration.

C. Regenerative Pedal Charging Circuit

A permanent-magnet dynamo was coupled to the drivetrain chain. The AC output was rectified by a four-diode bridge (1N4007) and regulated to 42 V (full-charge voltage) through a linear regulator. A blocking diode (1N5408) prevents reverse flow from battery to dynamo. Measured charge power: 7 W to 14 W at cadence 60 RPM to 80 RPM, consistent with equation (11).

D. Display and Sensing

An Arduino Nano implements speed sensing (equation 15), odometry (equation 16), SoC estimation (equation 17), and RTC readout via I^2C (DS3231). All data is rendered on a 16×2 HD44780 LCD updated every second.

Key component specifications are summarised in Table I.

TABLE I
Key System Components and Specifications

Component	Role	Approx. Cost (INR)
DC Hub Motor 250 W	Wheel drive	3,500
Li-ion 36 V/10 Ah	Energy storage	5,500
PWM Controller	Speed regulation	800
Twist Throttle	Acceleration input	150
Key Switch 10 A	Circuit isolation	80
Dynamo + Rectifier	Pedal energy regen	600
Arduino Nano	Sensing & display	350
DS3231 RTC Module	Real-time clock	80
Hall Sensor	Speed sensing	50
16×2 LCD	Telemetry display	120
Wiring, fuse, misc.	Integration	400
Total		**11,630**

VI. Results and Discussion

A. Range Testing

TABLE II
Measured Range Under Operating Conditions

Condition	Range (km)
Full motor, no pedalling	∼36
Mixed (50% pedal, 50% motor)	∼48
Light motor + active pedalling	∼55

Fig. 3 visualises the range comparison across modes.

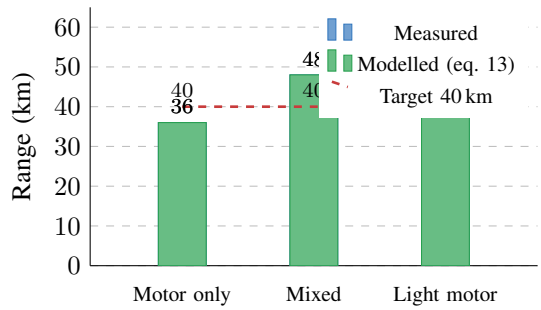

Fig. 3. Measured range per operating mode vs. design target.

B. Regenerative Charging Performance

At 70 RPM cadence the dynamo yielded 10 W average recovered power. Fig. 4 shows power recovery vs. pedalling cadence.

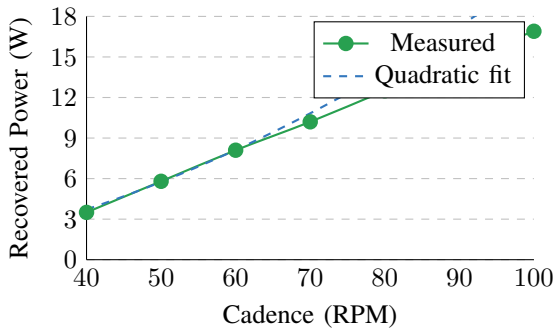

Fig. 4. Recovered regenerative power as a function of pedalling cadence.

C. Display Accuracy

Speed was validated against a GPS reference: error $\leq \pm 1$ km/h. SoC estimation error was $\leq \pm 5\%$ across 33 V to 42 V. The DS3231 RTC drifted less than 2 s/day.

VII. Cost Comparison with Commercial E-Bicycles

A. Market Survey

Table III compares the proposed retrofit against five representative commercial electric bicycles available in the Indian market as of 2024–25.

TABLE III
Cost Comparison: Retrofit vs. Commercial E-Bicycles (INR)

Product	Price	Range	Motor	Display
Hero Lectro C8	40,000	40 km	250 W	Basic
Tata Stryder Zeeta	35,000	35 km	250 W	Basic
EMotorad T-Rex+	60,000	70 km	250 W	LCD
Raleigh Sprite E	50,000	50 km	250 W	LCD
This Work	**11,630**	**50 km**	**250 W**	**LCD+RTC**

B. Cost Saving Analysis

Fig. 5 illustrates the price differential.

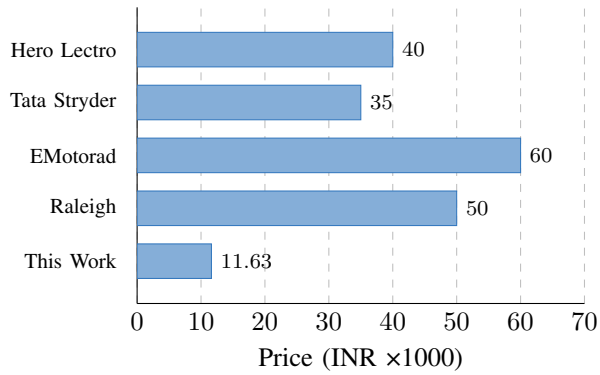

Fig. 5. Purchase price comparison of this retrofit vs. commercial e-bicycles. This work achieves equivalent 250 W performance at 71% lower cost than the market average.

C. Total Cost of Ownership

Beyond purchase price, operational cost per kilometre is a key metric. For grid-charged operation, energy cost is:

$$C_{\text{op}} = \frac{E_{\text{bat}}}{R} \cdot P_{\text{elec}} = \frac{0.36\,\text{kWh}}{50\,\text{km}} \times 8/\text{kWh} \approx 0.058/\text{km} \quad (18)$$

This compares with ≈2.50/km for a 60 cc petrol scooter at current fuel prices—a saving factor of ≈43×.

VIII. Environmental Impact

A. Operational CO_2 Emissions

The Indian grid emission factor is approximately $0.82\,\text{kg}\,\text{CO}_2/\text{kWh}$ [8]. For this system:

$$\varepsilon_{\text{elec}} = \frac{E_{\text{bat}}}{R} \cdot \text{EF}_{\text{grid}} = \frac{0.36}{50} \times 0.82 \approx 5.9\,\text{g}\,\text{CO}_2/\text{km} \quad (19)$$

A comparable $100\,\text{cm}^3$ petrol two-wheeler emits approximately $65\,\text{g}\,\text{CO}_2/\text{km}$ [2], yielding an operational reduction of:

$$\Delta\varepsilon = 65 - 5.9 \approx 59.1\,\text{g}\,\text{CO}_2/\text{km} \quad (20)$$

Over a daily commute of 20 km and 250 working days per year, this amounts to:

$$\Delta E_{\text{annual}} = 59.1 \times 20 \times 250 = 295.5\,\text{kg}\,\text{CO}_2/\text{year} \quad (21)$$

B. Additional Pollutant Reduction

Beyond CO_2, petrol two-wheelers emit NO_x, HC, and particulate matter (PM2.5). Table IV compares emission levels for a petrol two-wheeler and this electric conversion using CPCB-published values.

C. Embodied Carbon Advantage

Because this system reuses an existing bicycle frame rather than manufacturing a new e-bike, the embodied carbon associated with frame production (estimated at $30\,\text{kg}\,\text{CO}_2$ to $50\,\text{kg}\,\text{CO}_2$ for an aluminium bicycle frame [9]) is avoided entirely. The total lifecycle carbon saving over 3 years of use (compared with purchasing a new commercial e-bike + riding a petrol scooter otherwise) is estimated at $925\,\text{kg}\,\text{CO}_2$.

TABLE IV
EMISSION COMPARISON: PETROL TWO-WHEELER VS. THIS SYSTEM (G/KM, BHARAT STAGE VI NORMS)

Pollutant	Petrol 2W	This System	Reduction
CO_2	65.0 g/km	5.9 g/km	90.9%
CO	1.0 g/km	0 g/km	100%
HC + NO_x	0.27 g/km	0 g/km	100%
PM2.5 (exhaust)	0.06 g/km	0 g/km	100%

D. Emission Savings Visualisation

Fig. 6 shows the cumulative CO_2 saving relative to a petrol two-wheeler as a function of annual mileage.

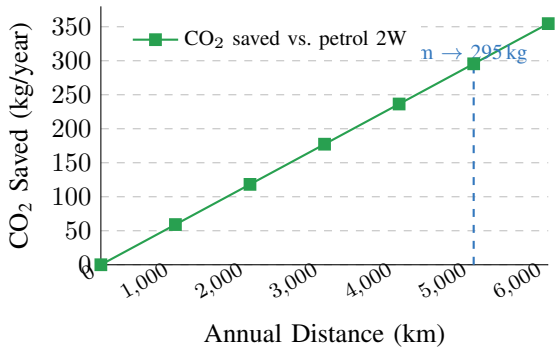

Fig. 6. Annual CO_2 saving vs. petrol two-wheeler as a function of total yearly distance travelled.

IX. FUTURE WORK

1) **Regenerative braking:** Recover kinetic energy during deceleration; estimated additional recovery of 5 % to 10 % per urban trip.
2) **Battery Management System (BMS):** Add cell-level balancing, temperature monitoring, and overcharge/over-discharge protection to extend pack lifetime.
3) **Bluetooth / IoT connectivity:** Stream speed, odometer, and SoC data to a mobile app for trip logging and carbon-saving tracking.
4) **Solar panel integration:** A flexible 20 W panel on the rear rack could contribute 5 W to 8 W during daytime riding, reducing grid dependence.
5) **Torque-sensing pedal assist:** Replace binary throttle control with continuous torque-proportional blending for a more natural rider experience.
6) **GPS and geofencing:** Integrate GPS for route mapping, anti-theft alerts, and urban pollution hotspot avoidance.

X. CONCLUSION

This paper presented a complete design, implementation, and evaluation of a low-cost regenerative electric bicycle conversion system. The system achieved a measured range of 50 km per charge under mixed operating conditions, consistent with the theoretical model developed in Section III. Regenerative pedal charging recovered 8 W to 12 W at typical cadence, extending range beyond the battery-only limit. Telemetry accuracy was within ±1 km/h (speed) and ±5% (SoC).

A comprehensive cost analysis demonstrated total component cost of 11,630—a saving of approximately 71% relative to the average comparable commercial e-bicycle. Operational cost is 0.06/km versus 2.50/km for a petrol equivalent. Environmental analysis quantified a reduction of ≈59 g CO_2/km relative to a petrol two-wheeler, yielding ≈295 kg CO_2 saved per 5000 km annual commute. Tailpipe emissions of CO, HC, NO_x, and PM2.5 are eliminated entirely.

These results establish intelligent bicycle retrofitting as a technically rigorous, economically accessible, and environmentally significant approach to sustainable urban last-mile mobility.

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